C.R.O.P.: A Farm-Facing Portal for Conservation and Revenue Opportunities

Raja Ali, Logan Harris, Cody Hatfield, Kanav Khurana, Tanay Patel, Ethan Perdieu, Rohit Rao, Amith Revelly, Abhinav Vellalacheruvu

PURDUE UNIVERSITY | The Data Mine

INDIANA SOYBEAN ALLIANCE® | INDIANA CORN Marketing Council

INTRODUCTION

***C**onservation and **R**evenue **O**pportunity **P**ortal*

$35.1B	94,350	15M	150+
Indiana ag economy	farmers across 15M acres	Indiana farmland	conservation & incentive programs

SCATTERED PROGRAMS
Programs are scattered across USDA, NRCS, ISDA, SWCDs, and private companies.

MISSED OPPORTUNITIES
Lack of a single access point means missed funding and lower program adoption.

LEGISLATIVE IMPACT
Indiana House Bill 1149 (2025) significantly increases the need for a solution.

ONLINE PORTAL

THE PROBLEM

Without C.R.O.P.:

- Search **dozens** of websites individually
- Different **formats**, **deadlines,** and **requirements**
- **No way to filter** by county, crop, or practice
- Programs go **unfilled**; **practices are NOT adopted.**

> *"I had no idea the NRCS would pay you."*
> — *Indiana farmer*

The Nature Conservancy · pepsico · USDA · indigo

OUR SOLUTION: C.R.O.P.

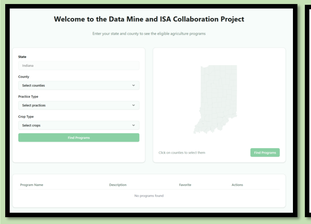


Fig 1: Landing Page.

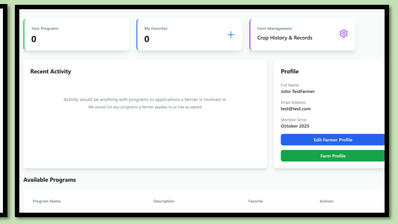


Fig 2: Farmer Dashboard.

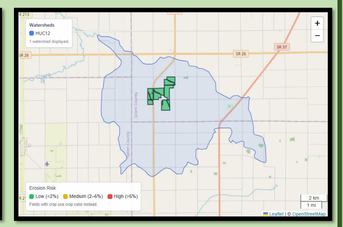


Fig 3: Field map with field & watershed boundaries with erosion risk via boundary file upload.

SCAN HERE ------>
TO
SEE IT IN ACTION!

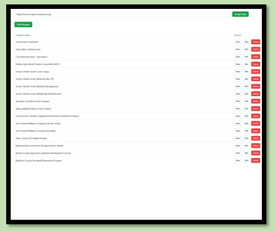


Fig 4: Admin Panel.

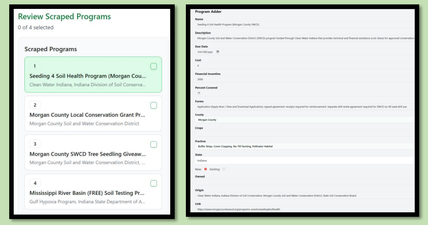


Fig 5 & 6: Scraped program information found automatically from one starting URL.

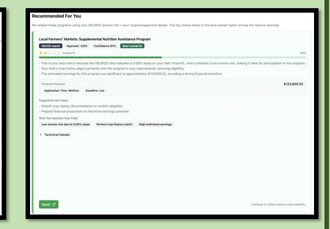
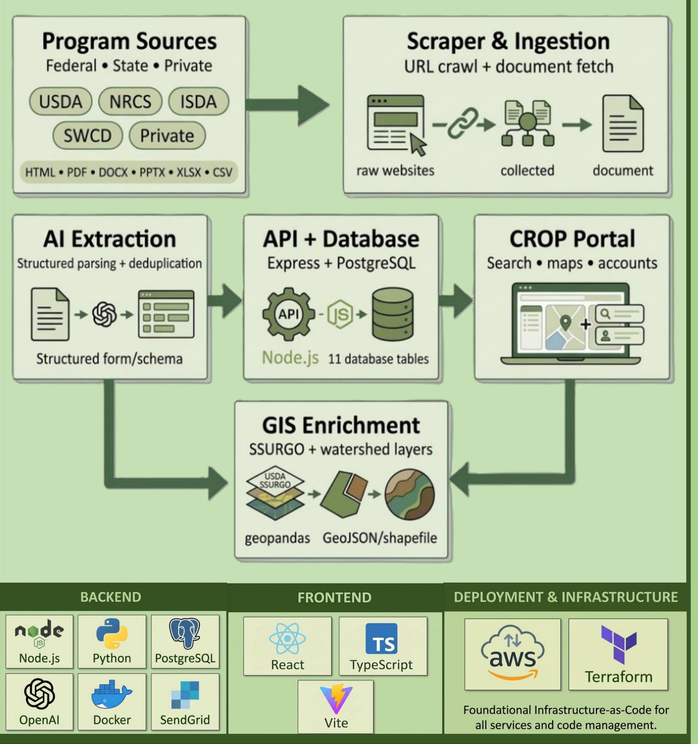


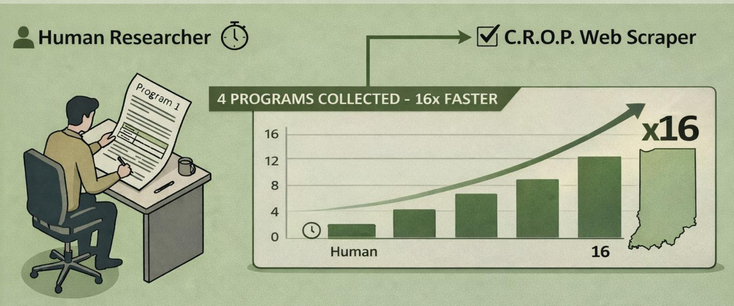
Fig 7: Program Recommendation.

METHODOLOGY



KEY RESULTS

C.R.O.P. WEB SCRAPER: 16x FASTER THAN HUMAN RESEARCHER



Manual Time (4 Programs): 37 min 28 sec

Scraper Time (4 Programs): 2 min 18 sec

MASSIVE TIME SAVINGS FOR FARMERS

Secure Farmer Platform
Email verification
Password reset
reCAPTCHA
Access & Refresh Tokens

Farm & GIS Tools
Field dashboards
Shapefile upload
SSURGO analysis
Erosion mapping
Leaflet Visualization

Data, AI & Export
URL Scraping & 6 file types
Automated data extraction
CSV/Excel export
AWS deploy
CI/CD automation

FUTURE GOALS & CONCLUSIONS

Advanced Analytics Dashboard
A new dashboard to measure farmer engagement and track the program's overall impact.

Autonomous Application Assistant (AI Agent)
Our AI Agent will guide farmers step-by-step through their application processes and workflows.

Automated Reminders & Notifications
Implement a system for sending automated deadline alerts and program milestone notifications.

Data Collection & Portal
Expand the centralized portal using automated data extraction from diverse government websites.

Enhanced Program Discovery
Allow farmers to search and filter by: **County, Crop Type**, and **Conservation Practice.**
Supports Indiana House Bill 1149 (2025)

IMPACT: **C.R.O.P. simplifies access to conservation funding by turning scattered government programs into a single searchable platform for farmers**

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